

School of Psychology
Information Sheet



**The University of
Nottingham**

UNITED KINGDOM • CHINA • MALAYSIA

Title of Project

LANDOLT – Asymmetry in visual cortex responses to dark and light stimuli [pilot stimuli for neuroarchitecture / visual calm]

Ethics Approval Number: F1718

Researcher(s): Visual Neuroscience Group (Psychology), colleagues at the SPMIC

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This is an invitation to take part in a research study that aims to improve our understanding of visual, sensory processing by using behavioural experiments and brain imaging.

A substantial body of research provides evidence for an asymmetry in the visual processing of light and dark elements in an image. In particular, stimuli darker than the average background (e.g. black dots, letters, stimuli) are processed more efficiently than lighter ones, with threshold improvements of up to 25% in some tasks (Lu & Sperling, 2012). Intriguingly, this asymmetry can change in early and mid stages of macular degeneration, a retinal disorder that is associated with loss of central, high-fidelity vision.

We will use functional magnetic resonance imaging (fMRI) in combination with carefully calibrated visual stimuli and a task to measure differences in the response to ON (light on gray) and OFF (dark on gray) stimuli across identified visual areas.

[Towards the end of the experiment, we may also show you images or short videos of scenes and measure if brain responses differ to indoor and outdoor settings and those corresponding to calmer and more busy working environments.]

Before you decide if you wish to take part, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully.

What will it involve for you?

If you participate, you will be asked to complete a brief medical questionnaire and to sign a consent form. You will then be asked to attend the Sir Peter Mansfield Imaging Centre, for a scan usually not lasting more than 1 hour on a visit. On some occasions we may ask participants if they could participate in follow-up visits at a later date.

You will be shown the scanner and we will explain what you will experience while you are in it. Once you are in the scanner – in a comfortable position lying on your back – and we will make some initial measurements that allow us to prepare the rest of the session. You may hear different sounds during this period.

During the experiment proper, we will record the properties and activity of your brain while you are looking via a mirror at a series of images (“visual stimuli”) that are presented to you a computer display. In most cases, you will have to perform a simple task, for example pressing a button whenever the colour of a small square during to keep you alert. Alternatively, we may ask you to make judgments about the appearance of images or patterns displayed on the screen, which you will report by pressing one or more buttons. In each case, the person conducting the study will explain beforehand what will happen and the length of time that you will be scanned for.

In other kinds of experiments, we may also use tactile (touch) stimuli or auditory (sound) stimuli. These are delivered through a device that can apply gentle mechanical vibration stimuli to the skin (usually on the palmar surface of your hand). Sound stimuli, if applicable, will be delivered through a headset that is compatible with the MRI scanner.

For the whole procedure you will normally spend no more than 1h inside the brain scanner.

Are there any risks of MR scanning?

Magnetic resonance imaging (MRI) uses radio waves similar to those used in radio and TV transmission. These have a much lower energy than X-rays and as such are considered biologically safe. We scan at field strengths of up to

3T. We will be following strict national safety guidelines which are designed to prevent the potential hazards of MRI which are burns and electric shocks. Preventable accidents like skin burnt at the contact point have never occurred in the Sir Peter Mansfield Imaging Center (SPMIC) and have only very rarely occurred elsewhere in the UK. While there is no evidence to suggest that MRI is harmful during pregnancy, the Medicines and Healthcare products Regulatory Agency (MHRA) advises against scanning pregnant women in the first three months of pregnancy or at fields above 2.5T. If you could possibly be in the first 14 weeks of pregnancy (e.g. if you have had unprotected sexual intercourse since your last period) then you must not be scanned. Pregnancy tests are available in the women's toilets of the SPMIC.

What will we do if we notice something abnormal?

Since you are healthy, it is extremely unlikely that your scan will show any abnormality. Even if there were an abnormality it is unlikely that we would notice it since we are taking these scans for scientific research, so they are not the same as scans collected by doctors for medical purposes. Furthermore, the pictures will not be looked at by a radiologist (a doctor qualified to find abnormalities in scans).

However, there is a small chance that we will notice something abnormal on your scan. This might have the benefit of allowing you to start treatment earlier than you would have otherwise, but it may also have implications for your future ability to find employment and obtain insurance.

If we did find anything abnormal on your scan the investigator would arrange for an appropriately qualified doctor to look at them. That specialist doctor would contact your GP to explain the situation, so that your GP could then advise you.

You will be asked to complete a standard MRI safety questionnaire and a form relating to tattoos.

Participation in this study is totally voluntary and you are under no obligation to take part. You are free to withdraw at any point before or during the study.

All data collected will be kept confidential and used for research purposes only. It will be stored in compliance with GDPR.

If you have any questions or concerns please don't hesitate to ask now. We can also be contacted after your participation at the above address.

If you have any complaints about the study, please contact:

Standard Information Sheet Version 1 2015

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